

FIELD VERIFICATION WORKSHEET

Asset: CV-101 - Garage conveyor (Conv_Simple) As of: 2026-07-11 Technician: _____

SAFETY: LOTO + wait >= 5 min for DC-bus discharge before touching.

SAFETY: A monitored e-stop is NOT a safety stop - verify the physical safety circuit removes drive power per NFPA 79 / EN 60204-1 before relying on it.

ID	LOCATION	WHAT TO CHECK	EXPECTED	MEASURED	PASS FAIL	INITIALS	DATE
-- E-003 --							
OI-14	E-003	Trace conductors on the bench; verify Q1 poles 1-6 actual locations. R-C surge absorber both ends recommended per GS10 manual.	6 a		P / F		
OI-15	E-003	Identify the bench supply; confirm CB1 device and rating. NEC and GS10 manual require upstream protection.	record as found		P / F		
OI-16	E-003	Photograph GS10 nameplate and motor nameplate; measure PE resistance between drive/motor grounds and supply PE.	record as found		P / F		
OI-17	E-003	Determine bench PE bonding strategy; verify RFI jumpers per GS10_UM.txt L1693-1718 if present.	record as found		P / F		
OI-21	E-003	Confirm whether a separate motor contactor exists between CB1 and VFD1; the device labeled 'MLC' (Q1) is a Schneider TeSys CA3KN22BD CONTROL RELAY - it has no main power poles.	1 a		P / F		
OI-27	E-003	Closed by technician confirmation - review/PHOTO_EVIDENCE_V7.md §3 ('it's 230'), not PHOTO_EVIDENCE_V6.md (which predates this confirmation and still shows 480 V/pending). Exact GS10 model/frame still tracked under OI-16 (for wire-gauge/fuse tables).	480 V		P / F		
-- E-003, E-005, E-006 --							
OI-19	E-003, E-005, E-006	Read the actual wire markers on the panel; replace proposed W-numbers with as-built (mirror of OI-03).	record as found		P / F		
-- E-004 --							
OI-25	E-004	Trace/meter the distribution block conductors against PS1 +V/-V output; confirm color-to-polarity mapping, circuit count, and any branch fusing (photo wire_1/wire_2). Separately, identify the branch circuit feeding PS1.L/PS1.N and its voltage (single leg of the 230 V 1phi supply, or a dedicated 120 V circuit).	230 V		P / F		
OI-28	E-004	Meter PS1 enclosure/DIN rail to the PE bus for continuity; meter DB1 0V-bus to the PE bus for continuity/resistance; confirm whether a deliberate 0V-PE bond exists or the 24 VDC system is intentionally floating.	0V		P / F		
-- E-005 --							
OI-01	E-005	Meter PS1 +24V to 0V = 24 VDC; confirm which terminal feeds the input devices.	24V		P / F		
OI-02	E-005	Confirm which embedded input common serves I-00..I-05 and whether inputs are sink or source; meter COM0 to 0V.	05 a		P / F		
OI-03	E-005	Read the actual wire markers on the panel; replace proposed numbers with as-built.	record as found		P / F		
OI-04	E-005	Confirm each device's real terminal markings against its nameplate/datasheet.	record as found		P / F		
OI-05	E-005	Confirm whether S0 lands directly on I-02/I-03 or via a safety relay; a monitored input is NOT a safety stop (see safety note).	record as found		P / F		
OI-06	E-005	Identify the physical sensor; confirm BN/BU/BK to +24V/0V/I-05; PNP sourcing assumed.	24V		P / F		
OI-07	E-005	Confirm selector type and that FWD/REV contacts map to I-00/I-01.	record as found		P / F		
OI-08	E-005	Confirm no field wires land on I-06..I-11 (program shows none).	record as found		P / F		

-- E-005, E-006 --

ID	LOCATION	WHAT TO CHECK	EXPECTED	MEASURED	PASS FAIL	INITIALS	DATE
OI-13	E-005, E-006	Open the live Conv_Simple_1.8 CCW project and cross-check every I-xx and O-xx assignment against the drawn functions (I-05 drift already proven; O-02 has independent Prog_init corroboration).	record as found		P / F		
<i>-- E-006 --</i>							
OI-09	E-006	Resolve per 2080-IN009 + meter the bank feed (expect PS1 +24V/0V).	24V		P / F		
OI-10	E-006	Meter coil circuit from O-02.	record as found		P / F		
OI-11	E-006	Meter the light circuits and confirm device markings.	record as found		P / F		
OI-12	E-006	Trace the PLC output block and verify no wires to O-04, O-05, O-06.	record as found		P / F		
OI-18	E-006	Trace the PLC output block; confirm zero hardwired leads to any GS10 digital inputs (fallback path should be unpopulated).	record as found		P / F		
OI-26	E-006	Trace each aux contact to its destination in the control circuit; confirm function (e.g., interlock feedback, seal-in, indicator).	record as found		P / F		
<i>-- E-006, E-007 --</i>							
OI-22	E-006, E-007	Map which of FWD/REV/DI3/DI4/DI5/AI/AO1/DO1/DOC are landed and to what; reconcile hybrid Modbus + hardwired-terminal control.	record as found		P / F		
<i>-- E-007 --</i>							
OI-20	E-007	Fresh keypad readback of the GS10 serial config; update print if params differ from current 9600/8N1 assumption.	1 a		P / F		
<i>-- panel --</i>							
OI-23	panel	Confirm whether the Siemens CPU 1212C controls any CV-101 function before referencing it on any CV-101 sheet; not drawn until evidenced.	record as found		P / F		
OI-24	panel	Identify the device (network device / meter?) and its relationship, if any, to CV-101.	record as found		P / F		

Field verification completed by: _____ Date: _____